

CHAPTER 2

Blockchain: a new perspective in cyber technology

T. Venkat Narayana Rao¹, Purva Pravin Likhar², Muralidhar Kurni³ and K. Saritha⁴

¹Department of Computer Science and Engineering, Sreenidhi Institute of Science and Technology, Hyderabad, Telangana, India; ²Mahatma Gandhi Institute of Technology, Hyderabad, Telangana, India; ³Department of Computer Science, School of Science, GITAM (Deemed to be University), Hyderabad, Telangana, India; ⁴Independent Researcher, Ananthapuram, Andhra Pradesh, India

1. Introduction

Development is an inevitable condition of humankind. With each day passing, millions of people find themselves in the continuous process of searching for solutions to billions of problems that require global attention. The exchange of cash has been a long-standing issue. The rise of banks and other intermediary people like brokers has contributed to cash exchange, making it a more complex problem to ensure legitimate transactions between any two parties.

It was 2008 when a solution finally surfaced to put the problem at hand to rest, as shown in Fig. 2.1. *Bitcoin* soon enjoyed huge success and became the most famous cryptocurrency, with its market reaching billions of dollars in less than a decade after its invention (Damak, 2018). To facilitate the transactions involving Bitcoin, a special data storage structure called *blockchain* was designed, and soon enough, this technology was used in other spheres (Frizzo-Barker et al., 2020).

1.1 Blockchain technology and its history

As the saying “Rome was not built in a day” goes, blockchains were not invented and developed in the 21st century. This technology dates to 1982 when Cryptographer David Chaum proposed a protocol similar to blockchain (Zhang et al., 2019). A few years later, Stuart Harbor and Scott Stornetta described a “cryptographically secure” chain of blocks in 1991, and their work concentrated on timestamping documents. They aimed to design a system that would maintain the integrity of these documents. In 1992, *Merkle trees* were incorporated into the design, which facilitated collecting several documents into a single *block*, improving the system’s

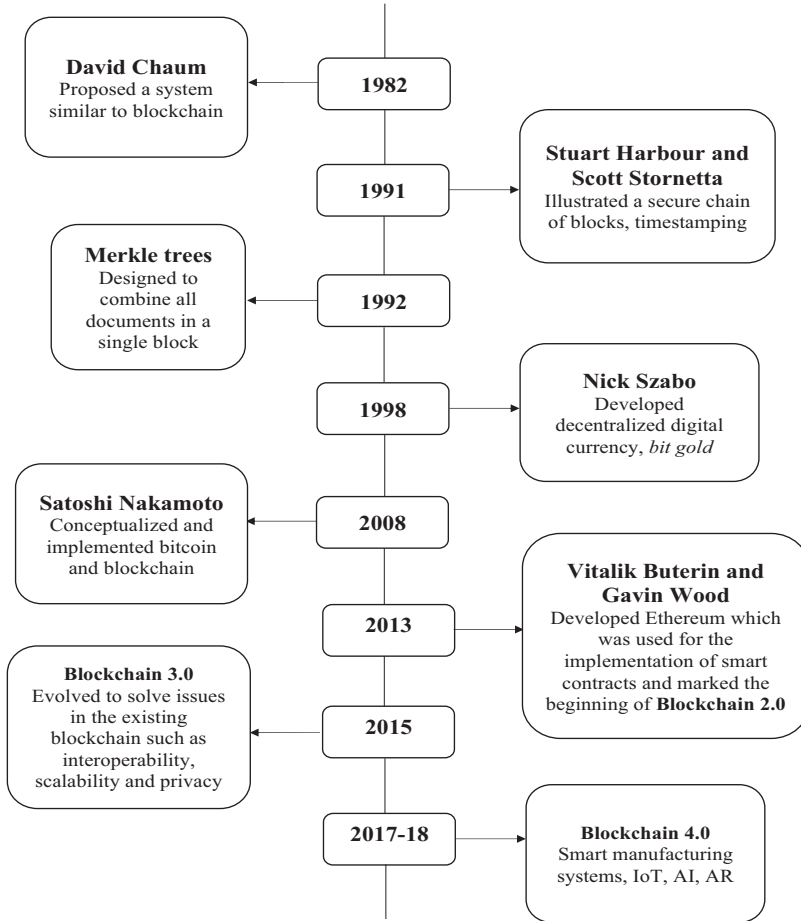


Figure 2.1 Blockchain technology evolution.

efficiency (Yangui et al., 2019). Digital currencies came into the picture in 1998 when a renowned computer scientist, Nick Szabo, started working on and designing something called *bit gold*, decentralized entities (Nian and Chuen, 2015). Finally, in 2008, the theory of blockchains and bitcoins was conceptualized and deployed under the name Satoshi Nakamoto, an anonymous developer or community of developers (Caradonna, 2020). Nakamoto authored the Bitcoin white paper and designed a system that used hashing techniques to timestamp blocks without a third party's involvement. It is speculated by many that Nick Szabo is, in fact, Satoshi Nakamoto since *bit gold* is said to be a direct precursor to Bitcoin architecture.

While it was not the first time an online currency was proposed, Bitcoin solved problems beyond imagination. It could be used as a form of virtual cash that could be sent peer to peer without a bank or other authorities and assisted the ledger's maintenance by an unbiased party. The engine that runs this ledger is known as "blockchain," and the first block of Bitcoin ever mined, the *genesis block*, was created by Nakamoto in 2009, and this block is known to be the foundation for all other blocks in the Bitcoin blockchain (Forsstr, 2015). As years passed, blockchain came into use for applications beyond currency and has been classified into four progressive eras. While Blockchain 1.0 was primarily focused on cryptocurrencies, Blockchain 2.0 included smart contracts, smart property, security trading, banking, and other finance areas. Blockchain 3.0 marked the technology usage in healthcare and biomedicine, arts, culture, science, and technology. The era of Blockchain 4.0 allowed decentralizing of IT systems and expanding these on cross-business processes like supply chain management, asset and resource management, and financial management (Xu et al., 2019).

2. Blockchain architecture

Blockchain refers to a sequence of blocks that contains a list of all immutable transactions occurring in a distributed network of nodes avoiding a single point of failure and denies the empowerment of a central authority to control its working. It is also referred to as a public distributed ledger system. This means that each node on the network possesses a copy of the shared data, so if any one of the nodes attempts to alter the data, it will not go unnoticed.

Fig. 2.2 represents a simple blockchain with a sequence of three blocks (Ismail and Materwala, 2019). As can be seen, each block contains a parent block hash. This hash value contains the identity of its previous block. If any blocks are removed, replaced, or misplaced with other blocks in the chain, they can be easily identified by verifying with this value. It is to be noted that the genesis block does not have a parent block, so the parent block hash value equals 0.

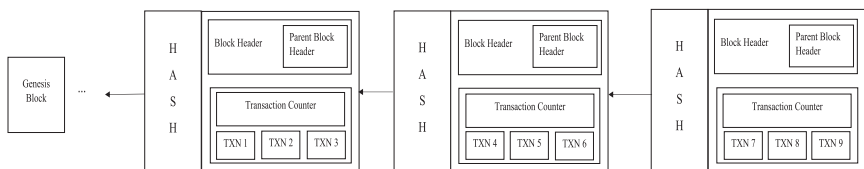


Figure 2.2 Simple blockchain.

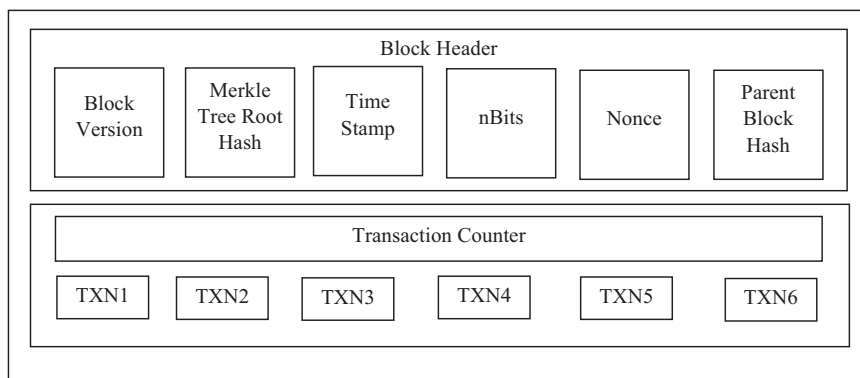


Figure 2.3 Block structure.

Fig. 2.3 represents the structure of a typical block. Each block consists of two main parts: *block header* and *block body*.

2.1 Block header

The block header contains the following fields:

- a. **Block version:** This indicates which set of block validation rules or protocols are to be followed.
- b. **Merkle tree root hash:** It is the hash values of all transactions present in the block body.
- c. **Timestamp:** It provides the current time in seconds since January 1, 1970.
- d. **nBits:** This is the target threshold of a valid block hash.
- e. **Nonce:** It is a 4-byte field that starts with 0 and increases with each hash calculation made.
- f. **Parent block hash:** As discussed earlier, it is a 256-bit hash value that points to the previous block.

2.2 Block body

A block body comprises two fields: the transaction counter and the transactions *TXN*. Depending on the size of each transaction's block and size, the maximum number of transactions may vary (Fig. 2.4).

To validate the transactions, blockchain makes use of asymmetric cryptography or public-key cryptosystems. These mechanisms make use of a digital signature to verify the authenticity of transactions. The typical digital signature algorithm used in the blockchain is the elliptic curve digital signature algorithm. For example, when a user *A* wants to send a message to another user *B*, the following process occurs (Fig. 2.5).

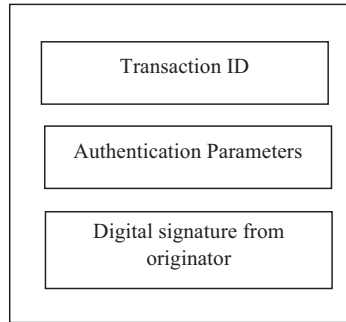


Figure 2.4 Transaction.

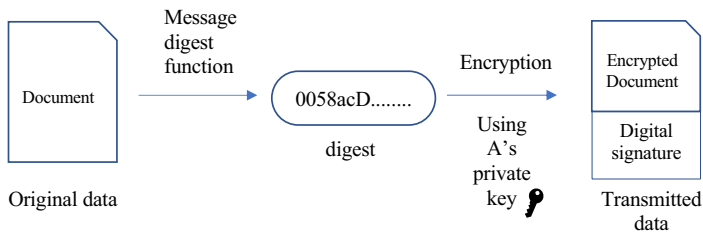


Figure 2.5 Signing.

On the original document, a message digest function is applied that creates a digest value, after which using the sender A's private key, the document is encrypted and a digital signature is attached.

After the receiver B receives the document, the same message digest function is applied that was applied on the originator end to obtain a digest value. Using B's public key, the transmitted document is decrypted and another digest value is obtained. If these values are the same, the signature is verified and added to the transactions block, as shown in [Fig. 2.6](#).

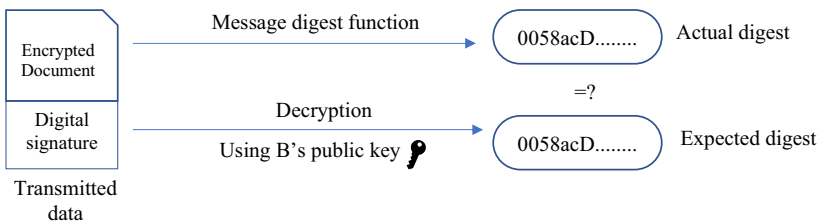


Figure 2.6 Verifying.

It is important to note that an additional database is also maintained by all the nodes known as the *Unspent Transaction Outputs* cache (UTXO) aside from the blockchain. It is a ledger with the records of funds available for every address and acts as a cache for the blockchain. As new transactions occur, UTXO gets updated: funds from sending addresses are subtracted and added to the receiving addresses. The Bitcoin Node software uses a key-value store, LevelDB, to hold a copy of the UTXO.

3. Blockchain concepts

Blockchain is a technology based on two essential concepts: *hashing* and *timestamping* (Sabry et al., 2019).

3.1 Hash functions

A hash function uses an algorithm that transforms an input of arbitrary length into a fixed-length hash value. Hash values generated are the same for the same input value. Hash functions map input to output, so small differences in input produce large differences in the result and are often used in computing since they locate data quickly using hash tables. Bitcoins use cryptographic hash functions, also known as secure hash functions, to perform consensus operations. These functions possess additional features over and above usual hash properties, such as the following:

- One-way function or resistance: If a hash value is given, finding the input data must be infeasible.
- Poor resistance to collisions: Given an input, finding another input with the same hash value must be computationally too tedious.
- Good resistance to collisions: It is computationally infeasible that they point to the same hash given two data points.

Bitcoin uses *SHA-2562* as its proof-of-work function (which will be discussed later in the chapter). SHA-2562 refers to the application of the hashing algorithm SHA-256 twice. As evident as it is, SHA-256 produces an output that is 256 bits long and meets the preimage resistance requirement (Fig. 2.7).

In the Merkle-Damgård construction, cryptographic hashing functions are built using a compression function as a building block to accept arbitrary length input, where the compression function refers to the scrambling of the input by a variety of shifting and mixing operations. Merkle-Damgård has shown that if the compression is collision-resistant, then the whole

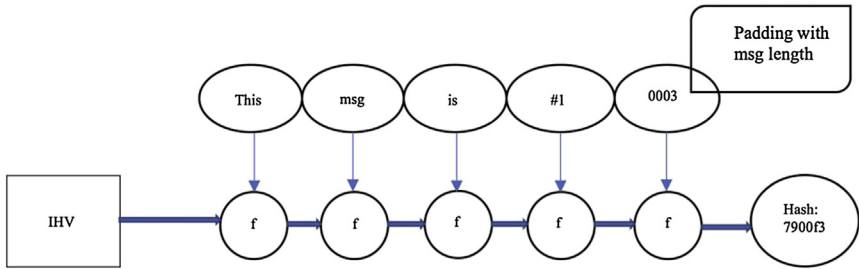


Figure 2.7 Merkle-Damgard construction.

construction is also. Referring to [Fig. 2.7](#), and with the compression function labeled f , it is to be noted that f operates on 256 bits in the case of SHA-256. The compression function takes two inputs: a block of input data or a message and an intermediate hash value. To compute the hash value using SHA-256, input data is broken into blocks of length 256 bits with the end of the message padded with zeroes. The intermediate hash value is given to IHV (initial hash value), and the compression value is applied to each block by applying the hash value from the previous step to the next step. Finally, the last value yields the whole input's SHA-256 hash.

The Merkle-Damgard structure is prone to *length extension attacks*. Ferguson et al. suggested that one way of fighting the length extension problem would be to hash the message twice, and it is believed that Nakamoto chose SHA-256² instead of SHA-256 for the same reason.

3.2 Timestamp

A digital timestamp is the same as a physical timestamp. It establish that some information, like a digital document, remained at a particular point in time, and it serves useful when two parties agree to a contract or transactions take place over a digital medium. Usually, the information in the timestamp is the hash of the data that is to be secured. This has several advantages:

- (i) The timestamp information is kept private and separate from the medium that is used to secure the timestamp.
- (ii) It decreases storage costs since a hash is considerably smaller than the information it generates.
- (iii) Digital signatures work best on data whose size is already determined.

There are many ways through which a timestamp can be secured. The easiest way would be to send a trusted party a copy of the information, and they would store this data and the time of reception in a safe place. The problem with such a method is that a trusted third party may lose the database or be compromised. Another way would be to rely on a *timestamping authority* (TSA), a trusted third party, to sign the data and the time when the data was communicated with its private key. This signature would be sent back to the original owner, but the timestamp will lose its integrity if the TSA collaborates with another party.

A third method would involve publishing the hash of the data in a public place, such as the newspaper. If an attacker wishes to alter any of the transactions, it will collide with the hash published in a public newspaper, as shown in Fig. 2.8. A TSA could use this idea too, but it would prove costly to publish a hash for every transaction. Instead, hashes of multiple data can be combined to achieve economies of scale. This can be executed efficiently by using Merkle trees, covered in Section 7.

Another refinement to further secure timestamps could be the usage of lined timestamps, as shown in Fig. 2.9. The main aim behind this is that if the hash to be published is connected to the hash already published, it will help protect the older hash. An attacker with malicious intent would now have to face a double collision if he/she desires to alter data from an older hash, and after another step in the chain, the collision would be triple and so on. Thus, the security of this chain increases exponentially.

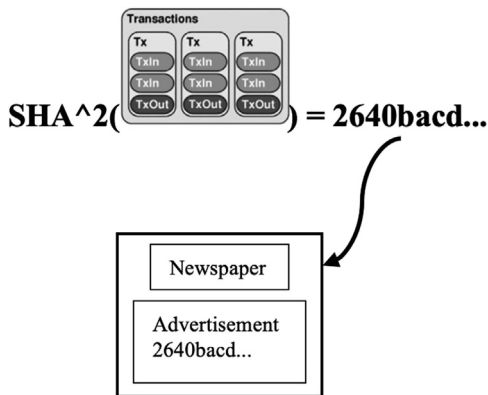


Figure 2.8 Timestamping transactions by publishing hash in newspapers.

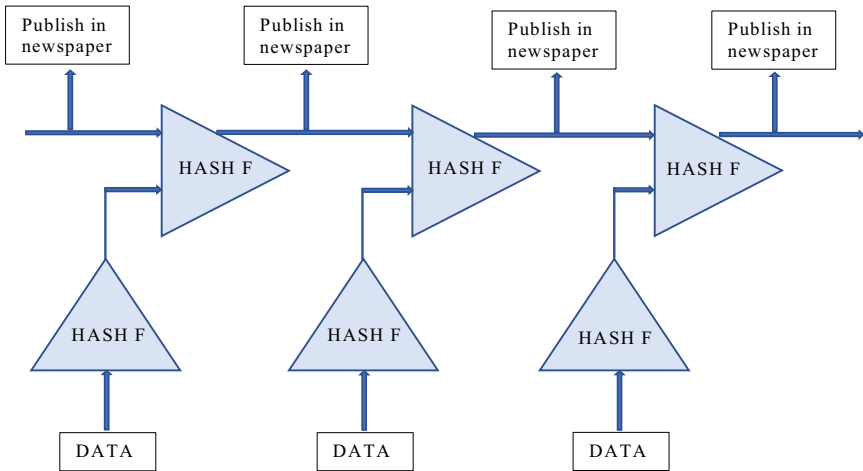


Figure 2.9 Linked timestamps.

Let us review some simple Java code to create a block to the blockchain.

```
// creating a block in a blockchain

import java.util.Date;

public class Block {
    private String hash;
    private String previoushash;
    private String data;
    private long TimeStamp;
    private int nonce;

    //constructor
    public Block(String data, String previoushash, long TimeStamp) {
        this.data = data;
        this.previoushash = previoushash;
        this.TimeStamp = new Date().getTime();
        this.hash = calcBlockHash();
    }

    //function to calculate hash
    public String calcBlockHash() {

        String calchash = crypt.sha256(previoushash +
        Long.toString(TimeStamp) + data);

        return calchash;
    }
}
```

In this code, four main attributes are seen that are used to create a block:

- *previoushash*: hash of the previous block
- *data*: actual content in the block, which may include numbers, strings, or a combination of both
- *TimeStamp*: the time of the creation of the block
- *hash*: the hash value of the current block
- *nonce*: used for cryptography

The method *calcBlockHash()* uses the user-defined class *crypt* to calculate the hash of the current block by concatenating *previoushash*, *TimeStamp*, and *data* values. This value is stored in a variable, *calchash*, which is then returned (Fig. 2.9).

Generating hash values:

```
// generating hash of the present block

public class crypt {

    public static String sha256(String input) {

        try {
            MessageDigest md = MessageDigest.getInstance("SHA-256");

            int i = 0;
            String hex;

            byte[] hash = md.digest(input.getBytes("UTF-8"));
            StringBuffer hexhash = new StringBuffer();

            while ( i < hash.length) {
                hex = Integer.toHexString(0xff & hash[i]);

                if (hex.length()==1)
                    hexhash.append('0');

                hexhash.append(hex);
                i++;
            }

            return hexhash.toString();

        }

        catch (Exception e) {
            throw new RuntimeException(e);
        }

    }

}
```

In this snippet, the following points must be noted:

- Using *MessageDigest*, an instance of the hash function SHA-256 is obtained.
- Then, a hash value for our input data, stored in a local variable *input*, is generated in the form of a byte array, *hash*.
- Finally, this byte array is transformed into a hexadecimal hash, *hexhash*, which is then converted into a string.

The principle behind the blockchain technology is mutual trust or consensus, and the algorithms in the next section are the last pieces of the puzzle to achieve it.

4. Consensus algorithms

Before diving into the algorithms, it is important to briefly understand the meaning and importance of consensus in blockchain technology.

4.1 What is the consensus?

The Byzantine Generals (BG) Problem occurred in the 13th century, which involved generals with their armies attempting to capture an encircled city. The attack would prove to be successful if all the armies attacked the city at the same time. Since all the generals belonged to different armies, coordination would be the main problem because their only way of communication would be through messengers. It is also possible that any of the generals may be traitors and may intercept the messages between two parties and attempt to modify them to cause the attack to fail. How to avoid single point failure, make a coordinated attack, and build trust among peers form the basis of today's blockchains.

Consensus refers to mutual trust or understanding between parties to agree on what is true. Consensus mechanisms or algorithms are strategies that a group of computers deploys in blockchains to achieve a common point of agreement or a single state of the network (Wang et al., 2019). There are many techniques to address this issue, which will be discussed further.

4.2 Consensus algorithm: Proof of Work (PoW)

In a decentralized network, a node is selected to record all the transactions within the network. The simplest way to select this node is to make a random selection, but its problem is that it is prone to attacks. Therefore, if

a node is interested in publishing a block of transactions, it must do much computational work to prove that it is unlikely to attack the network. This forms the basis of PoW. It is a strategy that deploys the nodes in the network to calculate the block header's hash. As discussed in the blockchain architecture, each block contains a nonce value, and the nodes would make changes to the nonce frequently to get different hash values. The consensus demands that the estimated value must be equal to or less than a specific given value. Once a node reaches this target value, it broadcasts the block to all the other nodes in the network, and all the nodes are supposed to confirm the accuracy of the hash value mutually. If it is validated, this new block is appended by all the nodes to their blockchains, and this process takes place every 10 min. The nodes contributing their computing ability to the partial hash inversion power are termed miners, and this process is called mining. In simple terms, mining is the process of guessing a nonce that generates a hash with the first x number of zeroes.

The following program aims to make the mining concept clear and use a brute force approach to achieve this goal.

```
//mining the block

public String miningblock (int prefix) {
    String prefstr = new String (new char[prefix]).replace('\0',
        '\0');

    while (!hash.substring(0, prefix).equals(prefstr)) {
        nonce++;
        hash = calcBlockHash();
    }
    return hash;
}
```

In the code, the following process is seen:

- Define the desired prefix.
- Then, check whether the solution has been obtained.
- If the solution has not been found, increment the nonce and calculate the hash using *calcBlockHash()*.
- This process continues until the required result is attained.

[Fig. 2.10](#) illustrates the PoW working.

The question is *why would anyone want to go through a complex problem and solve it computationally and keep the blockchain growing?* This is because miners

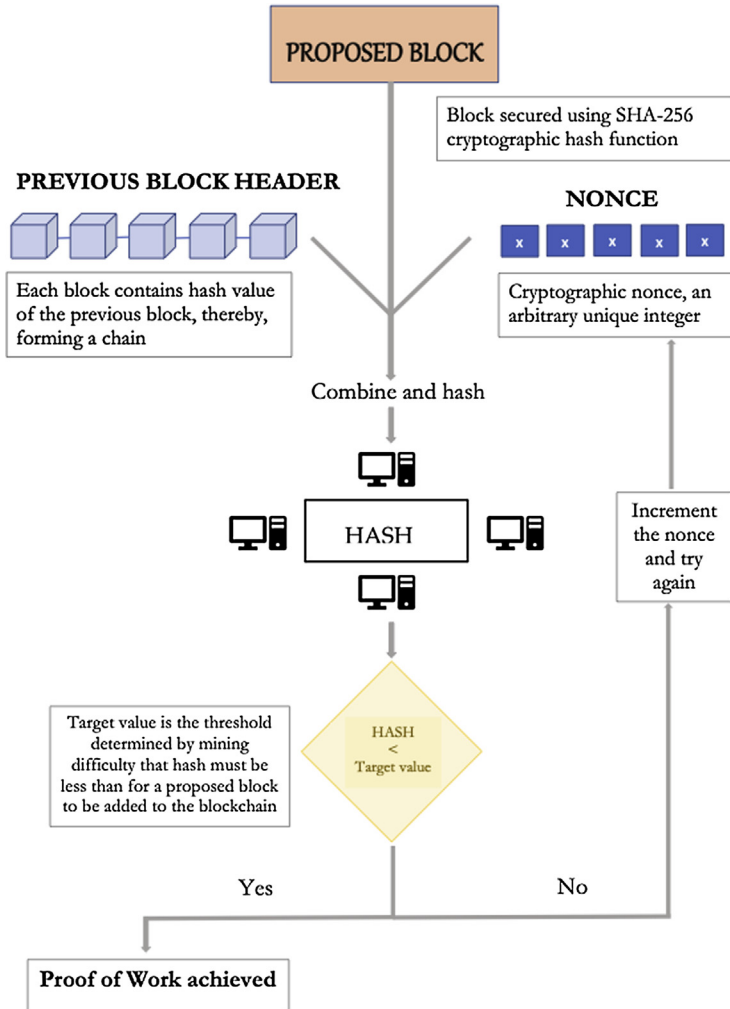


Figure 2.10 Proof of Work.

receive *mining rewards* in the form of coins or any other way that the application may want to reward its participants. The winning miner is given the block reward and the transaction fees every 10 min.

Sometimes, two miners may arrive at the same block by finding the nonce at the same time. This may result in *forking*. This results in network division, and some miners regard one branch as valid, while others follow

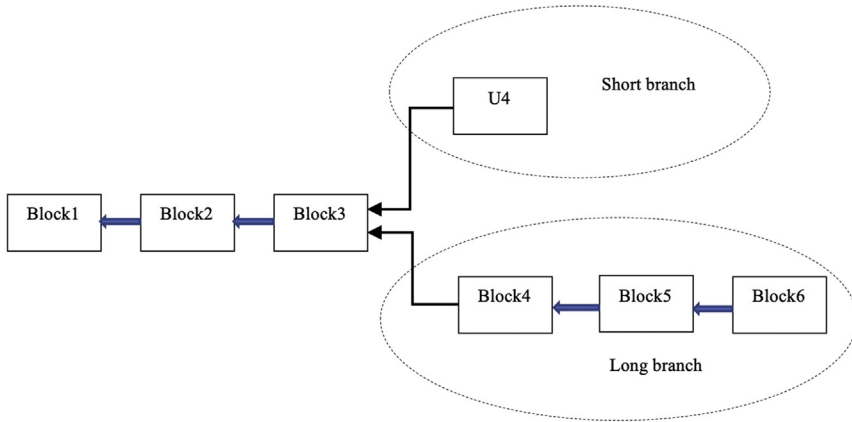


Figure 2.11 Forking.

the different branch of execution. The procedure specifies that the right branch is the longest, as seen in Fig. 2.13. Therefore, the other branch, *U4*, is discarded (known as *orphan blocks*), and miners stop working on it.

The problem with PoW is that computing calculations waste many resources. To mitigate these issues, some PoW protocols have been designed to include a side application.

4.3 Consensus algorithm: proof of stake (PoS)

PoS is an alternative to PoW, which locks crypto assets to protect the network. Miners in PoS are asked to prove their possession of the currency. The basic concept of PoS is focused on a randomly chosen state of validators who “stake” the native network tokens by locking them into the blockchain to generate and accept blocks. Validators are compensated directly proportional to their total stake, incentivizing nodes to validate the network based on a return on investment. Then, based on their stakes, validators are picked to follow their next blocks. While often programmed with random functions to prevent front-running consensus, a greater amount staked by a validator gives them a higher probability of producing the next block. Proposed blocks by validators are then propagated or broadcasted to the rest of the set, verifying and appending the blockchain’s accepted block, as shown in Fig. 2.11.

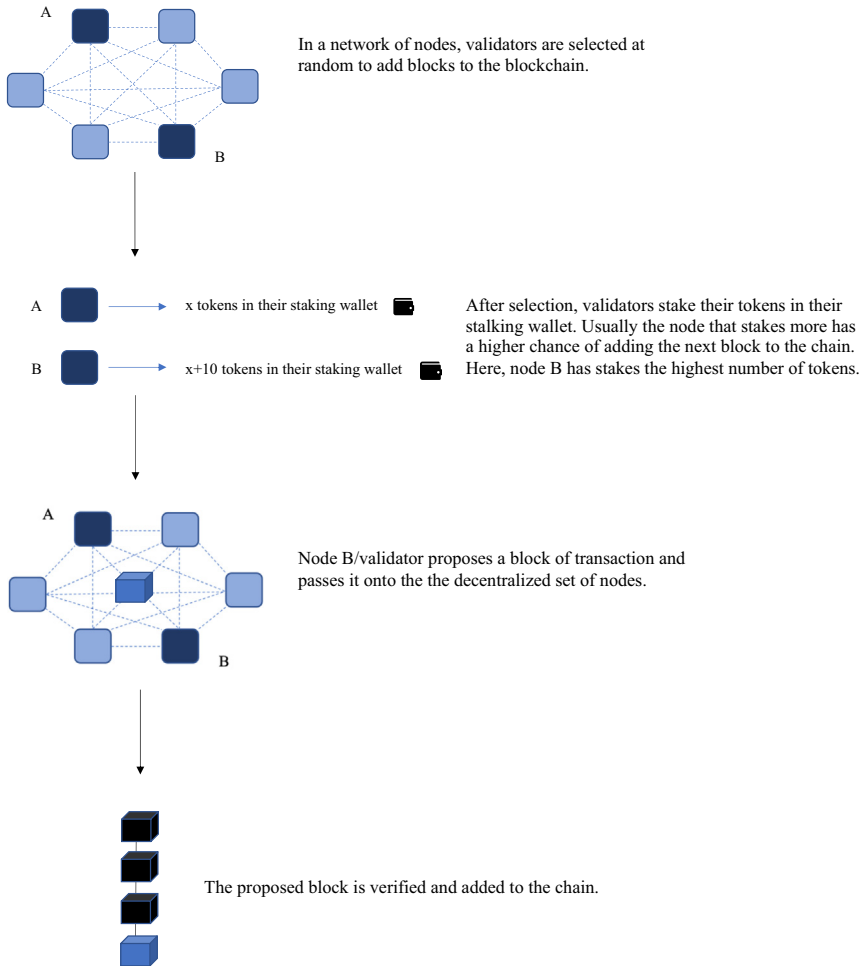


Figure 2.12 Proof of stake validators.

It is usually believed that people with greater amounts are less likely to attack the network, but this gives dominance to the richest person. Therefore, many solutions were identified with a combination of the stake size to decide which one will forge the next block, like the lowest hash values, the age of the set of coins, etc. Many blockchains adopt PoW initially and ultimately transform into PoS, like the one Ethereum has been

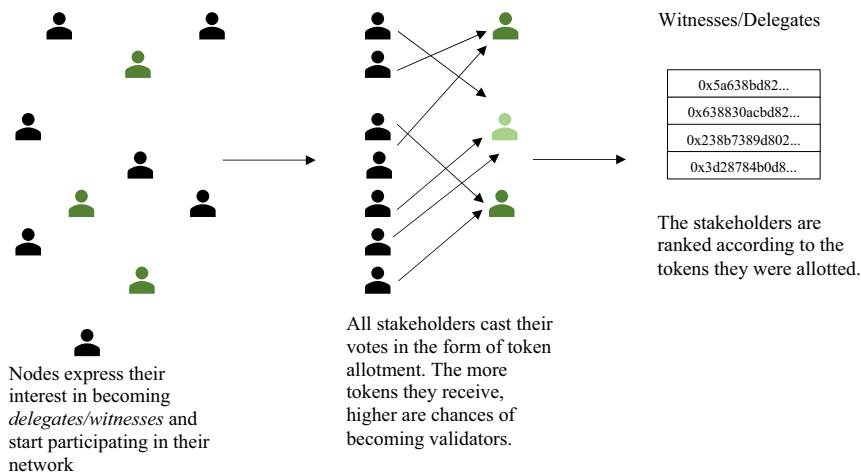


Figure 2.13 Delegated proof of stake.

developing. This kind of hybrid model is known as *Casper*. Unlike in PoW, miners in PoS are not rewarded with a mining reward. Instead, they charge a transaction fee or a network fee to add blocks to the blockchain.

4.4 Consensus algorithm: delegated proof of stake (DPoS)

The significant difference between DPoS and PoS is that DPoS is representative democratic. In the sense that stakeholders elect their delegates to generate and validate their blocks, they appoint them as validators. Since the nodes are fewer in number, validation takes less time, so transactions are made faster. Additionally, delegates are free to tune block size and block intervals, and users may vote dishonestly to delegate if needed (Figs. 2.12 and 2.13).

4.5 Consensus algorithm: Practical Byzantine Fault Tolerance (PBFT)

PBFT is a replication algorithm proposed to tolerate *Byzantine faults*, the problems that occurred in the BG problem, and it is known to handle one-third of these faults. The whole process is divided into three parts: *preprepared*, *prepared*, and *committed*. In each round, a *primary* (0) is selected according to some set conditions and is responsible for ordering transactions; a client C

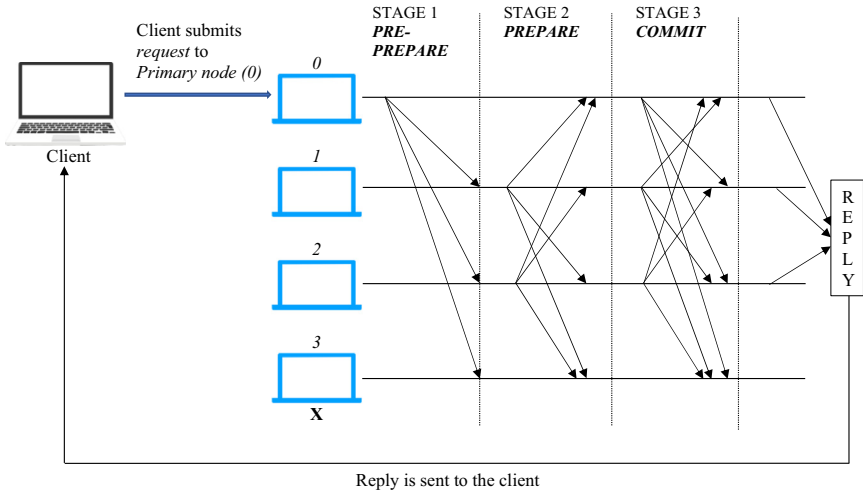


Figure 2.14 Practical Byzantine Fault Tolerance.

submits a request to the primary node. The primary sends messages to everyone (nodes 1,2,3, wherein the third node may have lost connection and dropped from the network). Other nodes accept the message as long as it is valid. Increasingly, these messages often contain signatures and sequence numbers to determine message validity (Fig. 2.14). If a node accepts a prepared message, it recognizes it by sending a prepared message to everyone. The receiving nodes accept these prepared messages as long as they are true. A node is then called prepared if it received the original request from the main, preprepared message and saw $2f$ prepare messages (received from other nodes), making it $2f+1$, where f is the Byzantine fault number. After nodes have been prepared, they send a commit message to the rest of the nodes, and if a node receives valid commit messages from $f+1$, they perform the client's request and send the reply to the client. On the other hand, the client waits for $f+1$ answers and declares them correct. A node would reach the next step if it obtained two-thirds of all nodes. PBFT demands that the network knows each node and queries other nodes.

Table 2.1 explains the comparison among the algorithms discussed based on the mentioned criteria.

Table 2.1 Consensus algorithms comparison.

Property Algorithm	PoW (proof of work)	PoS (proof of stake)	DPOS (delegated proof of stake)	PBFT (Practical Byzantine Fault Tolerance)
Energy saving	No	Partial	Partial	Yes
Tolerated	<25%	<51%	<51%	<33.3% faulty
power of	computing	stake	validators	replicas
adversary				
Application	Bitcoin	Peercoin	Bitshares	Hyperledger fabric
of algorithm				

5. Blockchain validity

Before diving into how to check if a block on a blockchain is valid, these blocks must be stored to be verified later. In the program below, these blocks are stored in an array list of *blockchain*.

```
//to store blocks of a blockchain, we make use of ArrayList

import java.util.ArrayList;

public class Storingblocks {

    //creating ArrayList object
    public static ArrayList<Block> blkchain = new ArrayList<Block>();

    public static void main (String[] args) {

        blkchain.add(new Block("1st block", "0"));
        blkchain.add(new Block("2nd block",
            blkchain.get(blkchain.size()-1.hash));
        blkchain.add(new Block("3rd block",
            blkchain.get(blkchain.size()-1.hash));
        blkchain.add(new Block("2nd block",
            blkchain.get(blkchain.size()-1.hash));
        blkchain.add(new Block("2nd block",
            blkchain.get(blkchain.size()-1.hash));
        blkchain.add(new Block("2nd block",
            blkchain.get(blkchain.size()-1.hash));

    }

}
```

```
//validate blocks on blockchain

public static Boolean ChainValidity() {
    Block presentblock;
    Block prevblock;

    for(int i=1; i < blkchain.size(); i++) {

        //storing the current and previous block

        presentblock = blkchain.get(i);
        prevblock = blkchain.get(i-1);

        //checking if present hash equals to calculate hash or not

        if (!presentblock.hash
            .equals(presentblock.
                calcBlockHash())) {

            System.out.println("Hashes are unequal");
            return false;
        }

        //checking if previous hash equals to calculated previous
        //hash value or not

        if (!prevblock.hash
            .equals(presentblock.prevhsh)) {

            System.out.println("Previous hash values are
            unequal");
            return false;
        }
    }

    //if all hashes are equal to their respective calculated hashes,
    //then it the blockchain is completely verified

    return true;
}
```

6. Blockchain attacks

A double-spend attack happens when two separate transactions try to expend the same funds. Usually, the protocol defends this attack by deciding the valid transaction as the one that makes its way to the blockchain. Thus, this issue is addressed in a decentralized manner without the intervention of a single central authority. A transaction is further made safe when more blocks are stacked on top of the block where it was first

included. Therefore, if an attacker wishes to alter a particular block, he will have to mine all the blocks to make way to the desired block (Moubarak et al., 2018). Since adding blocks is a continuous process, the attacker would have to redo all the precedent work and keep pace with the new blocks added to the blockchain. The only way to accomplish such a task is to command a hash rate as high as the other hash rates of the network. The hash rate refers to hash rates for blocks in the blockchain. This attack is known as a 51% attack, and an attacker who controls less than half of the network is still possible. The likelihood of success depends on the hash rate regulation of the attacker and the number of blocks to dig in, and this likelihood decreases exponentially. If the attacker can monitor >50% of the network hash rate, the likelihood of success becomes 100%. A 51% attacker can change transactions he controls and trigger double spending on those assets but cannot change the transaction receiver, as the elliptic curve cryptography (ECC) signature protects it.

A race attack happens by checking some nodes when a vendor accepts payment of an unconfirmed transaction. The attacker may submit a transaction to vendor-near nodes and a different transaction to several other network nodes. Thus, only the nodes closest to the vendor reveal the transaction sending funds to the vendor, while the remaining network contains double spending in their mempool. To defend against this attack, a vendor should wait for the transaction to be at least one block (Moubarak et al., 2018).

Bitcoin developer Hal Finney discovered the Finney attack. Moreover, a miner is required to premine one transaction into a newly created, unreleased block and spend the coins. If the payment is still unconfirmed, the new block will include a double-spending transaction. The solution to solve the Finney attack is to wait for at least six confirmations.

Attackers may also try to perform a DoS (denial of service) attack on the network by generating surplus transactions to fund themselves and stop legitimate transactions. However, this offense is unlikely to happen since the number of free transactions an attacker can send to the network is of a limited number, the fees to complete this payment would have to compete with the fees of a legitimate transaction, and these transactions must be made above a certain threshold.

7. Merkle trees

The data structure proposed by Ralph Merkle in 1982 is one Merkle tree or hash tree. Nakamoto assumed the concept of Merkle trees to timestamp

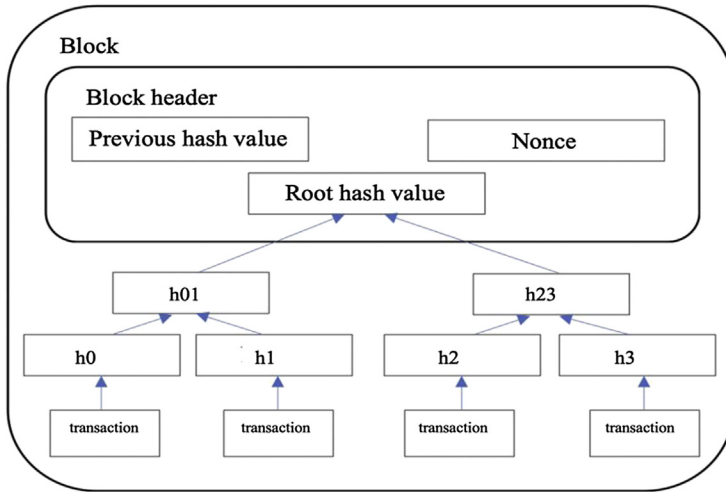


Figure 2.15 Merkle tree of transactions for a block.

a block of transactions. These trees are commonly used in file-sharing applications to keep track of blocks of files (Mercan et al., 2020). Fig. 2.15 shows how Merkle trees compute the hash value of a block. Initially, a binary tree, where each parent node has two children, is formed with hashes of individual transactions as leaves denoted by $h_0, \dots, h_n$. The hash of the parent node is denoted by the hash of its two children, i.e., $h_{01} = H(h_0, h_1)$, where H stands for SHA-256 hash function. Finally, the hash of the root node, Merkle root, is calculated, and now this can be appended to the block header (refer to Fig. 2.15).

Transaction authentication is one of the main drawbacks of the Merkle tree. Assume that a node would like to search whether a t_3 is a block transaction. A node can accomplish this task in $\log(n)$ time when n is the number of nodes. The node has to measure h_3 , h_{23} , and the root hash and verify the root hash stored on the block according to Fig. 2.15. This means the node can only search the unique branch of Merkle.

8. How secure is blockchain?

The original version of blockchain uses the SHA-256 algorithm, which was already discussed earlier in the chapter. Despite being secure, SHA-256 faced inefficiency issues due to which specialized hardware, *mining rigs*, were developed to handle the processing of transactions and confirm

and rely on valid transactions by solving mathematical problems called *hashes*. Apart from hashes, the cryptographic information on each block of records contains a unique timestamp, making the blockchain more secure (Bashir, 2018).

Imagine a person adding links to the chain of blocks, and someone comes along, chooses a link in the middle of the chain, and starts adding his blocks to the link. This results in forking (refer to Fig. 2.13). Blockchains can be forked for various reasons, like falsely altering a record on the preexisting chain, or someone may wish to take control of the whole system. It is simple enough to compare the hash and timestamp of the block to verify the original, valid record in both cases. This scenario happened in 2013 due to a faulty update, but the swift reaction by blockchain developers and warning miners showed that it is possible to recognize and isolate fraudulent forks quickly. An alert expert on the blockchain can isolate a node that has gone rogue or shows signs of tampering without compromising the rest of the system, which benefits from having a decentralized system.

9. Challenges and recent advances

Given the potential of blockchain, it faces quite a few challenges that hinder its use. Some shortcomings and advances in the areas are as follows (Chen, 2018; Zheng et al., 2017; Pajoooh et al., 2021).

9.1 Scalability

Blockchain becomes bulkier as the number of transactions grows daily. Each node must have the details of transactions to validate and verify on the blockchain. Due to the block and time interval's original size to generate a new block, blockchain can process only seven transactions per second. Moreover, because the block's capacity is less, small transactions might get postponed because miners prefer transactions with higher transaction fees. There were two methods designed to fight this problem:

- a. storage optimization of blocks by adopting a novel cryptocurrency scheme that ensures removal of old transactions by networks using *Ver-Sum*, which allows lightweight clients to outsource expensive computation over large inputs;
- b. redesigning blockchain through *Bitcoin-NG* (next generation), whose main idea is to divide a conventional block into a key block for leader election and microbook to store transactions.

9.2 Selfish mining

In this strategy, selfish miners keep their mined blocks without broadcasting, and this private branch would be revealed only when some conditions are met. The private branch would ultimately grow longer than the current public chain; all the miners in the network would admit it; therefore, the honest miners would be wasting their energy and resources on a useless branch, and selfish miners would be mining on their unpublished private chain without anyone to compete with. Therefore, selfish miners would be earning more. To fix this problem, Heilman proposed a novel approach for honest miners to decide which branch to follow with random beacons and timestamps. Another approach would be to generate and accept each block within a maximum interval of time. This approach, known as *ZeroBlock*, would disable selfish miners from achieving more than expected revenue (Li, 2019).

9.3 Privacy leakage

It is a lesser-known fact that blockchain cannot guarantee transactional privacy because the values of all transactions and balances for each public key are made publicly available. New research has revealed that a user's Bitcoin transactions can be connected to his personal information. A method was proposed to connect user pseudonyms to IP addresses even when users use firewalls to protect their identity. However, the origin of the transaction could still be found. Since then, multiple methods have come into the picture to solve this issue:

- a. **Mixing:** It refers to the strategy that provides anonymity to users by exchanging funds from multiple source addresses to multiple destination addresses, so intermediary nodes can be used to transfer data between two parties. For example, consider A with address a wants to transfer funds to B possessing address b . If it sends money to B directly, their relationship might be revealed. On the other hand, if A sends money to a trusted intermediary C with multiple inputs $c1, c2, c3$, etc., and multiple outputs $d1, d2, d3, b$, etc., deliver the money to B , it becomes hard to reveal the relationship between A and B . However, there may come a point where these intermediary nodes like C may behave dishonestly. Mixcoin devises a simple method to avoid such behavior. The intermediate node encrypts user information, including fund amount and transfer date, with its private key, and if

the intermediate node fails to send the money, anybody could verify that the intermediary cheated. This method, however, detects theft but fails to prevent it. Coinjoin depends on a central server to shuffle output addresses to prevent theft. An improvisation to this, CoinShuffle uses decryption mix nets for address mixing and shuffling.

- b. Anonymous: In Zerocoin, a zero-knowledge proof mechanism is used. The origin of payments is unlinked from their transactions, so graph analyses cannot be made. In Zerocash, zk-SNARKS (zero-knowledge succinct noninteractive arguments of knowledge) is leveraged, so the transaction amount and the coin values that a user possesses are hidden.

10. Emerging applications of blockchain

Blockchain technology has been in use mostly in the financial domain. Specific industries have contributed tools to help financial institutions and governments in monitoring the exchange of cryptocurrency. They detect frauds, money laundering, and compliance violation issues, helping the general public to invest their trust in blockchains. Thanks to the rise in Blockchain 2.0, Blockchain 3.0, and Blockchain 4.0, others have been trying to apply this technology in their field. To enhance their systems and improve their performance, conventional industries can use blockchain for their secure storage facilities. The following areas have benefitted from blockchain extensively (Li, 2019) (Bassi et al., 2020):

10.1 Blockchain and Internet-of-Things (IoT)

The past few years witnessed many evolutions, among which blockchains and the IoT happened to impact the world significantly. IoT has become a sensation worldwide due to its unique network of physically connected devices (which contain components such as chips, sensors, codes, software, and exchange data) via the internet. Since devices are connected via the internet, there lies a possibility that this exchange of information might face security and privacy issues. This is where blockchain technology can come into use. It is essential to ensure that the information flowing between the devices is safe and protected from unwanted parties. Many industries have developed software and hardware that are connected and operated via blockchain. Some benefits in this area include encryption of

ledger data, real-time distribution of information between blockchain-connected devices, and monetization of machines based on timestamps. Following are some of the IoT systems that use Blockchain in their day-to-day use:

- **Energy CPS:** Energy grid systems come under complex cyber-physical systems or CPS, which direct some of the main functions such as energy generation, distribution, and utility in a bidirectional way. The main aim of blockchain technology is to provide a secure environment for interactions in such systems.
- **Vehicular CPS:** Many smart transportation systems, including unmanned aerial vehicles and autonomous driving cars, have been invented to solve traffic and road safety issues. Security and privacy are main drawbacks, so blockchain technology has been in use to leverage these intelligent systems due to their decentralized and immutable characteristics.
- **Smart homes:** We see several inventions regarding sensor-based lights, electrical systems such as fans and door locks, and devices that help other home operations. It is necessary to ensure that the right party is in command to dictate these devices' actions. Blockchain technology can be used to provide the right users with control of their devices.
- **Internet of battlefield things:** IoT has also been used in defense situations under the acronym IoBT. Smart defense vehicles and combat equipment with sensors are used to enable battlefield operations. Blockchain technology can, therefore, help in conducting secure and reliable commands for IoBT.
- **Industrial IoT:** IIoT enables the integration of wireless components connected via the internet with processes that support industrial activities. These industrial operations comprise examination, monitoring, analysis, production, management, and distribution of items manufactured. These processes must be given equal importance to attain efficiency, safety, and quality, which is where blockchains are deployed.

10.2 Smart contracts

Smart contracts contain functions that consist of a set of instructions that are embedded with conditional statements. These instructions are self-executable, which means that the contracts execute the actions specified

under the condition if the condition is met. Blockchain-based smart contracts are like agreements between parties who are obliged by the terms specified. The agreement is automatically implemented on the fulfillment of the conditions. When a new block is supposed to be added to the chain, miners do the computation work, and once all the users verify it, it is concatenated to the chain. Thus, smart contracts work only if there is mutual consensus in the group, thereby stressing the network of node decentralization.

10.3 Blockchain in education

Due to its strong cybersecurity capabilities, blockchain technology has made its place extensively in educational industry usage. Although its application in education is still in its early stages, blockchain can be widely used to issue, validate, and share data. Since blockchain comprises a digitalized, decentralized, and secure ledger, one can have their educational records under their authority and share these credentials with interested parties. Documents such as e-transcripts, digital degrees, and e-certificates can be easily exchanged among individuals or groups, like Nespore, who developed a blockchain certificate sharing platform to ease this process. Since blockchain technology keeps a complete record of a student's profile in data blocks in sequential order by timestamps, managing and verifying data becomes comfortable with full transparency and authenticity. Data blocks old and new cannot be deleted or tampered with, so it acts as a fool-proof virtual infrastructure for storage of credentials and achievements indefinitely.

Furthermore, the ubiquitous learning or U-learning system was introduced, which equipped students with an engaging and collaborative learning environment with high security with blockchains, which facilitated web-based distance education. This reduces costs, procedures, and unnecessary paperwork in schools, colleges, and universities. Blockchain technology aims to end a high-cost, trustless, inefficient data management paper-based system and provide a secure, authoritative, transparent system, so applications like EduCTX that are blockchain-based higher education credit platforms have grown popular.

Even though blockchain offers many advantages to the academic sector, there are quite a few limitations. Since there are thousands of students whose records need to be maintained, educational organizations need to keep track of all these records. Due to the increased block size, the latency

of transactions increases, making the system slow and repetitive, thereby giving rise to scalability problems. In addition to this, deploying blockchain also includes infrastructure costs, computing power costs, and time costs. There is also a possibility of security issues rising due to constant system updating to add newer features. Despite these issues, blockchain technology has been applied in many educational processes such as the following:

- Admissions: Students can easily fill out various applications and other formalities of different institutions across the world.
- University library and related services: With the advent of blockchain, library authorities can easily monitor the movement of books and students who have borrowed books. Obsolete methods of recordkeeping such as register maintenance can now be discarded with systematic ledger technology.
- Alumni contact: All necessary communication with students is stored in the form of blocks, which helps develop a good rapport with the alumni.
- Nonacademic achievement detail maintenance: Apart from education, the details of a student's extracurricular activities can be maintained as a part of his/her records, thereby facilitating a complete record of a student's merits.
- Intellectual property protection: Since many researchers in a university may be working on similar projects, it is possible that two parties may collide over intellectual property rights due to plagiarism. The use of blockchain avoids such a hassle by publishing research content while keeping an eye on citations and reuse.

10.4 Blockchain in healthcare

Although the medical industry is one of the last to put into use the latest technology, it is safe to say that blockchains might offer some advantages to the sector. The utility of blockchains can be expanded to medical fields such as pharmaceuticals, genome medicine, neurosciences, and clinical studies. Managing patient records so they are private and standardizing and looking for patterns among the huge, structured blocks of health data are main reasons for digitizing these details. Sensitive information related to patients and doctors must be transferred safely among parties without middlemen, and the accountability of all parties must be held. Due to blockchain-based smart contracts, the shared data parameters are maintained, and health plans are customized for patients. This would also help

improve the security and privacy of these records by handing over their access to patients, making them available and retrievable from any part of the world. The patient could then monitor their health with their complete data record, which can be used for reference elsewhere stored in the InterPlanetary File System (IPFS). Another advantage to the healthcare industry would be managing the whole process of drug manufacture and distribution. Most blockchain projects employed in hospitals are built on IPFS to utilize encryption and minimize data redundancy with the help of off-chain cloud components using crypto algorithms.

Various blockchain-supporting companies have started offering data monetization and tokenization services. For example, Nebula Genomics provides genome sequencing for free to stock a blockchain-based genetic marketplace. On sequencing their genomes, users can charge an amount to any person who wants to access it, and these tokens can be redeemed for future products and services. Another blockchain company named Genomes.io allows users to sequence their genomes and stores them securely, allowing access to only a select group of people. Users can sell their genetic data in parts and, at the same time, prevent their data from falling into the wrong hands. In collaboration with an fast healthcare interoperability resources (FHIR) enabled EHR (electronic health record) aggregator mobile application, Health Wizz allows users to safely aggregate, organize, exchange, and trade their tokenized medical data. This helps patients to exchange their data with organizations related to healthcare safely. Medicalchain, an HER blockchain company, allows doctors and other healthcare workers and healthcare-related institutes like hospitals, dispensaries, and pharmacies to access patient EHRs.

Since blockchains are decentralized, immutable, trustable sources that are robust and secure, several points can benefit a health industry:

- Improved medical record management: Unalterable and source-verifiable patient records are the prime advantages of having blockchains.
- Enhanced insurance claim processes: Eliminating middlemen with transparent blockchain records and detecting fraudulent insurance claims will become easier since records could be verified easily and checked if the claim can be qualified.
- Accelerated clinical and biomedical research: Since the patient data is timestamped and accurate, the right authority can use these records for research purposes.
- Advanced healthcare data ledger: Manufacturer and ownership transferring, drug detection, and prevention systems in pharmaceutical supply chains add more positive points of using a blockchain.

10.5 Blockchain in supply chains, transportation, and logistics

From the advent of blockchain, it has been used in many applications, including supply chain management (SCM). SCM refers to the management of a product from its point of origin to the point of consumption. These affairs must be handled with much care to maximize value and minimize time. Especially since the coronavirus pandemic hit, many things have been made digital. The most common problems that global SCM faces are tracking and tracing, opacity, and ambiguity. It has become essential to allow a solid link between consumers, their products, and values to maximize the profit that traditional systems have failed to offer. With the use of blockchains, it can be easy to identify the source of the final goods. This information can be easily verified since blockchains cannot be tampered with. Since records are timestamped, consumers can know the exact time at which each process in the chain took place, and due to transparency, the relationship between the producer and consumer is highly improved. The main factors that influence the application of blockchains into SCM are the following:

- trust
- efficiency
- cost-effectiveness
- authority and control
- sense of privacy
- scalability

The structural design of blockchains helps provide SCM resilience by identifying risks associated with any supply chain, mitigating the impacts of any disturbances caused in the supply chain, and providing encryption to data, thereby safeguarding it. Since security is a significant concern, blockchains help enhance IoT-based intrusion detection where product bar codes can ease the interaction between various components in the process and seamless communication using RFID (radio frequency identification) protocol. Smart contracts are used to verify and allow actions by the devices, and any action outside the scope of this application is not carried out. Shipping giants like Maersk teamed with IBM and Walmart have infused RFID systems to trace consumer products. Shipping firms like DHL have also found their application in blockchain by using a digital ledger of shipments, the progress of their shipped goods, and maintenance of the integrity of transactions. In areas of product management, blockchains have had a more significant role to play. Every day, tons of goods are produced,

and our obsolete systems are incapable of storing such information. Blockchains ensure that the exact pricing from raw materials to the completed goods is made known to the public.

Real-world projects like originChain have started to restructure their traditional databases to blockchains to provide tamper-proof traceability of imported products. While blockchain commercialization looks promising, the supply chain and logistics areas are still far from unlocking the true potential of blockchains in their fields.

10.6 Blockchain in the agri-food sector

The distribution of agricultural products had faced many issues in earlier times. Thanks to blockchain technology, the food being circulated across various levels is provided with security due to its transparency. In addition to this, since blockchains are low-cost, fast, and robust, they can be easily applied to the agricultural supply chain. Since each block is unique, identifiers can be assigned to each product, which would be then added to the block, and this is where the IoT can have a significant role to play. These identifiers could contain information about the type of product, its source, its cost, its expiry date, and the way it was distributed. Due to its immutable nature, food transactions could be scrutinized at each point, thereby promoting easy sharing of food details.

The trade of agricultural items involves many parties, i.e., the producers, processors, packers, traders, wholesalers, retailers, and consumers. It is possible that since information is passed through so many levels, problems related to synchronization among various stakeholders might rise and might lead to differences in content about the same product. Blockchains have paved a way to solve this problem by integrating all the data of different levels into a single block, allowing symmetry of information throughout. Furthermore, the food labels could be easily verified using their identifiers to prevent swindles.

AgriBlockIoT is a blockchain-based system that uses Ethereum and Hyperledger Sawtooth implementations, which allows the traceability of food using IoT devices. As discussed earlier, RFID technology has found its use in agri-food sectors too by tracing the items using simple components like labels and tags that contain related information stored in databases.

10.7 Blockchain in technology and cybersecurity

The integration of cloud platforms with blockchain is an essential step toward solving the technology sector's most significant problems since

clouds and IoTs are incredibly prone to threats due to their centralized nature. The entire process can now be automated and therefore avoid single point of failure issues.

Various organizations throughout the world find it challenging to share their threat information since there is a possibility that competitors may use the information to break into their systems. However, information can be shared easily when there is a two-key pair since only the public key is released publicly and the private key is known only to authorized members of the blockchain. With the usage of decentralized credential solutions, cybersecurity risks in IoT devices have seen a drop. Usage of biometric and password-free solutions instead of a password being stored on a centralized system has been trending and has made IoT devices almost virtually unhackable.

10.8 Blockchain in identity management

Blockchain technology has found extensive usage in identity management in the following forms:

- **Certificates:** Records of important certificates for birth, marriage, or death, etc., and other kinds of registrations such as work permits and pensions must be kept safe and secure. These details must last a long time, so traditional storage structures must be avoided to eliminate the risks of alteration. Such issues can be resolved with the use of public ledgering systems such as blockchain. Another adaptation to this can be to encrypt these certificates.
- **Academic records and transcripts:** Blockchains can store educational records and memorandums of students by keeping them verifiable and immutable.
- **Property rights:** In the entertainment industry, originalities are the only valuables for artists. The original work of the artists and their rights must be preserved. This may seem like a challenge to create an accurate database to allot rights. Blockchain technology and smart contracts prohibit fraudulent people from claiming someone else's work as theirs, thereby allowing ownership to the right party.
- **Citizenship rights:** Citizenship identification information such as passports and social security numbers is now digitalized, and the information is stored safely in blockchains. Owners of these can now be verified from anywhere in the world since digitalized passports are now available to government authorities. Blockchains may also benefit the voting system by allowing voters to cast a vote securely from anywhere in the world.

- Personal identity: Blockchains can be used to safeguard users' identity by encrypting their data and securing it from attackers. A person's identity is stored in a digital form that can be accessed from anywhere, replacing traditional physical card identifications.

11. Conclusion and future scope

This chapter discussed the history and events that led to the emergence of Bitcoin and blockchains, the architecture of blocks, different kinds of consensus algorithms, blockchain attacks, Merkle trees, blockchain security, and challenges.

Each algorithm has been known to serve various needs, and a developer has to pick the one that meets his/her needs the best.

- PoW is known for its simplicity and resistance to a wide range of cyberattacks.
- PoS is a more energy-saving alternative to PoW and reduces the complexity of the issues at hand by quite a significant rate.
- Due to the limited number of nodes, the DPoS can manage high volume transactions and has fast confirmation rates.
- PBFT is energy efficient, does not need multiple confirmations, and provides low rewarding variance.

Due to its decentralized design, blockchain is extremely resilient and completely transparent and has found its application in many sectors apart from finance and will continue to be a powerful force, as discussed in this chapter. This technology has made it a point to cross all its limitations and show a potential direction, thereby changing the whole dynamic around artificial intelligence.

The potential areas where blockchain can make unpredictable advancements are big data analytics, decentralization, and blockchain testing.

11.1 Big data analytics

Since blockchain focuses primarily on security and distributed storage, it could be used for data management in the coming years. It could also validate the authenticity of the data. This case would significantly help the health sector where patient data must be kept untampered and protected from theft. In the data analytics domain, transactions on the blockchain can perform operations on big data. Since data is huge and may predict a pattern, analytics could forecast stock prices and trading.

11.2 Decentralization

Blockchain was designed to be a decentralized system. However, miners in the process are centralized. It is estimated that the top five mining pools collectively own more than 51% of the Bitcoin network's total hash. To prevent selfish mining, blockchain may develop methods to solve this problem.

11.3 Blockchain testing

Since blockchain has come into use in different fields, users have begun to include them in their business, so it has become necessary to test which kind of blockchain would suit the users' requirements. This has given rise to blockchain testing mechanisms. A typical blockchain testing mechanism can be divided into two phases: *the standardization* and *testing phases*. During the standardization phase, the blockchain is created and agreed upon by a set of criteria to be fulfilled, and in the testing phase, a different criterion is set on which blockchain testing needs to be performed.

Apart from the consensus algorithms discussed, there are newer algorithms like Proof of Authority (PoA), Proof of Elapsed Time (PoET), Proof of Burn (PoB), and Proof of Capacity (PoC), a few of which are still in the developing stage, and the rest have been just finding their ground so they go beyond the scope of this chapter.

References

- Bashir, I., 2018. Mastering Blockchain. Packt Publishing, Birmingham. www.packtpub.com.
- Bassi, M.A., Lopez, M.A., Confalone, L., Gaudio, R.M., Lombardo, L., Lauritano, D., 2020. Blockchain: emerging applications and use cases. In: Nature, vol. 388, pp. 539–547.
- Caradonna, T., 2020. Blockchain and society. Informatik-Spektrum 43 (1), 40–52. <https://doi.org/10.1007/s00287-020-01246-7>.
- Chen, X., 2018. Blockchain challenges and opportunities: a survey Zibin Zheng and Shaoan Xie Hong-Ning Dai Huaimin Wang. IEEE Int. Symp High Perform. Distrib. Comp. Proc. 14 (4), 352–375.
- Damak, M., 2018. The Future of Banking: Cryptocurrencies Will Need Some Rules to Change the Game. S&P Global. <https://www.spglobal.com/our-insights/The-Future-of-Banking-Cryptocurrencies-Will-Need-Some-Rules-to-Change-the-Game.html>.
- Forsstr, S., December, 2015. Blockchain Research Report.
- Frizzo-Barker, J., Chow-White, P.A., Adams, P.R., Mentanko, J., Ha, D., Green, S., 2020. Blockchain as a disruptive technology for business: a systematic review. Int. J. Inf. Manag. 51 (November), 0–1. <https://doi.org/10.1016/j.ijinfomgt.2019.10.014>.
- Ismail, L., Materwala, H., 2019. A review of blockchain architecture and consensus protocols: use cases, challenges, and solutions. Symmetry 11 (10). <https://doi.org/10.3390/sym11101198>.

- Li, Y., 2019. Emerging blockchain-based applications and techniques. *Serv. Oriented Comput. Appl.* 13 (4), 279–285. <https://doi.org/10.1007/s11761-019-00281-x>.
- Mercan, S., Cebe, M., Tekiner, E., Akkaya, K., Chang, M., Uluagac, S., 2020. A cost-efficient IoT forensics framework with blockchain. In: *IEEE International Conference on Blockchain and Cryptocurrency, ICBC 2020*, April. <https://doi.org/10.1109/ICBC48266.2020.9169397>.
- Moubarak, J., Filiol, E., Chamoun, M., 2018. On blockchain security and relevant attacks. In: *2018 IEEE Middle East and North Africa Communications Conference, MENACOMM 2018*, pp. 1–6. <https://doi.org/10.1109/MENACOMM.2018.8371010>.
- Nian, L.P., Chuen, D.L.K., 2015. Introduction to bitcoin. In: *Handbook of Digital Currency: Bitcoin, Innovation, Financial Instruments, and Big Data*, April, 5–30. <https://doi.org/10.1016/B978-0-12-802117-0.00001-1>.
- Pajooh, H.H., Rashid, M., Alam, F., Demidenko, S., 2021. Hyperledger fabric blockchain for securing the edge Internet-of-Things. *Sensors (Switzerland)* 21 (2), 1–29. <https://doi.org/10.3390/s21020359>.
- Sabry, S.S., Kaittan, N.M., Ali, I.M., 2019. The road to the blockchain technology: concept and types. *Period. Eng. Nat. Sci.* 7 (4), 1821–1832. <https://doi.org/10.21533/pen.v7i4.935>.
- Wang, W., Hoang, D.T., Hu, P., Xiong, Z., Niyato, D., Wang, P., Wen, Y., Kim, D.I., January 2019. A survey on consensus mechanisms and mining strategy management in blockchain networks. *IEEE Access* 7, 22328–22370. <https://doi.org/10.1109/ACCESS.2019.2896108>.
- Xu, M., Chen, X., Kou, G., 2019. A systematic review of blockchain. *Financial Innovation* 5 (1). <https://doi.org/10.1186/s40854-019-0147-z>.
- Yangui, S., Bouguettaya, A., Xue, X., Faci, N., Gaaloul, W., Yu, Q., Zhou, Z., Hernandez, N., 2019. In: Y. N., E., Steffen, B. (Eds.), *Service-Oriented Computing – ICSSOC 2019 Workshops*. <https://doi.org/10.1007/978-3-030-45989-5>.
- Zhang, R., Xue, R., Liu, L., 2019. *Security and Privacy on Blockchain*, vol. 1. ArXiv, 1.
- Zheng, Z., Xie, S., Dai, H., Chen, X., Wang, H., 2017. An overview of blockchain technology: architecture, consensus, and future trends. In: *Proceedings – 2017 IEEE 6th International Congress on Big Data, BigData Congress 2017*, pp. 557–564. <https://doi.org/10.1109/BigDataCongress.2017.85>.